PURE PATH LAB — LABORATORY MANAGEMENT SYSTEM

Software Requirements Specification (SRS)

Single-lab pathology + USG / X-ray platform — registration, billing, reporting, barcode & QR verification, business accounts, and patient OTP access.

Version 1.1 • 03 October 2026 • Issued for acceptance

FIXED-PRICE BUILD	SUPPORT	STATUS
Rs 60,000	30-day defect-fix window AMC Basic **Rs 2,000** / month	Issued for acceptance Confidential
Recurring hosting, domain and messaging costs billed separately at actuals.		

STACK	HOSTING	DATABASE	OTP
React (Vite) + Express REST + MongoDB	Vercel (client account)	Atlas Free (M0)	MSG91 (~1,000/mo)

This document is the **single contractual baseline** for scope, price, exclusions, and acceptance of the Pure Path Lab build. It is derived from the live codebase (**frontend/** React, **backend/src/** Express, Mongoose models) and the commercial terms agreed on 03 October 2026. Anything not listed in §5–§7 is out of scope (§12) and will be separately estimated.

Document control

Field	Detail
Version	1.1 (baseline v1.0 revised with agreed terms dated 03 October 2026)
Stack	React 18 + Vite • Express 4 REST • Mongoose 8 • MongoDB Atlas
Hosting / DB	Vercel (client-owned) • Atlas Free tier M0
Build price	**Rs 60,000 (fixed price)**
Support	30-day defect-fix window • AMC Basic Rs 24,000/yr @ Rs 2,000/mo (agreed)
Status	Issued for acceptance

Version history

Ver	Date	Change
1.0	03 October 2026	Initial SRS from codebase audit (scope, commercials, dependencies)
1.1	03 October 2026	Agreed terms incorporated: domain, infrastructure ownership, MSG91 volume, DLT dependency, WhatsApp exclusion, language scope, print sizes, support and AMC terms

Contents

01 1 — Introduction
02 2 — Overall description
03 3 — Tech stack & architecture
04 4 — User roles & access
05 5 — Functional requirements (FR-1…FR-12)
06 6 — Patient portal (OTP login)
07 7 — Notifications (SMS / Email)
08 8 — Non-functional requirements
09 9 — Deployment & hosting (client-borne)
10 10 — Commercials: Rs 60,000 fixed
11 11 — Recurring / third-party costs (excluded)
12 12 — Out of scope
13 13 — Assumptions & dependencies
14 14 — Acceptance criteria (sign-off checklist)
15 15 — Agreed terms & client dependencies
16 16 — Support (AMC) & risk notes
17 Sign-off

Reading guide

Items marked **'to be completed'** are delivered within the Rs 60,000 build. Items marked **'as-is / minimal'** are retained in their current working state, with no further build-out in this scope.

1 Introduction

1.1 Purpose

Define **what will be built, delivered, and charged** for Pure Path Lab: a single-lab pathology + basic radiology (USG / X-ray) management system covering patient registration, billing, report entry → verification → print/PDF, QR verification, barcode labels, business accounts, and patient report access via Mobile + OTP.

1.2 Scope

- **In scope:** §5–§7 as implemented in **frontend/** + **backend/src/**, finished to a production-usable state and deployed per §9.
- **Single lab only.** No SaaS/multi-tenancy, subscriptions, trials, paywalls, or multi-branch (existing branch/modality scaffolding stays minimal/as-is).
- **Out of scope:** §12.

1.3 Definitions

Term	Meaning
Case / Bill	One billing event; auto-generates linked lab report(s)
TAT	Turnaround time in hours (**tatHours**)
Verify / Sign	Pathologist sign-off that finalises a report
WABA	WhatsApp Business Account — excluded from this phase

2 Overall description

2.1 Product perspective

Full-stack web application: a Vite React SPA served as a static build, backed by an Express REST API (**/api/***, serverless entry **api/index.js**), MongoDB Atlas for data, multer-based uploads for images/signatures/report files, and server-side PDF generation (pdfkit) + QR codes (qrcode) for verifiable reports.

2.2 Operating environment

- Desktop-first responsive web app; patient portal mobile-friendly.
- Online-only. Printing on Laser / Inkjet (A4) plus configurable sticker stock.
- Browsers: latest Chrome / Edge / Firefox.

3 Tech stack & architecture

Layer	Technology
Frontend	React 18, React Router v6, Axios, Vite build (**frontend/**)
Backend	Node + Express 4, JWT, bcryptjs, multer, pdfkit, qrcode (**backend/src/**, **api/index.js**)
Database	MongoDB via Mongoose 8 — **Atlas Free tier (M0)** at launch
Hosting	**Vercel** — Vite build to **frontend/dist**, **/api/*** rewrites per **vercel.json** (client account, separate cost)
Auth	JWT (staff) • Mobile + OTP sessions (patients, **OtpSession**)
Integrations	MSG91 OTP/SMS (client-funded) • Email via SMTP / generic send API • WhatsApp channel excluded

As-built API surface (extract): auth, users, patients, bills, reports, tests, usg, xray, modalities, doctors, doctor-portal, appointments, transactions, expenses, dashboard, analysis, export, notify, public / patient-portal, signatures, settings / setup, support-tickets, jobs, audit-log, branch, inquiry.

Core entities: Patient, Bill, BillItem, Report, Test, TestCategory, TestPanel, TestPackage, Interpretation, USGCase, XrayCase, ModalityCase, Doctor, Agent, Transaction, Expense, LabProfile, Signature, TatConfig, NotificationTemplate, DeliveryAttempt, OtpSession, SmsCredit, User, SupportTicket, Activity / Job.

4 User roles & access

Role	Access
Superadmin / Admin	Everything: settings, lab profile, users & roles, rates, accounts, verification
Pathologist (canSign)	Verify / sign / reject reports; signatures
Technician / Lab staff	Registration, billing, sample collection, result entry, image upload, print
Radiologist / USG / X-ray	Modality cases, image upload, findings, sign-off
Accountant / Cashier	Billing, collect / due / void, cashbook, expenses
Patient (no staff login)	Mobile + OTP → view / download own reports & bills

Enforcement: **User.permissions** map + server-side role checks; frontend permission matrix mirrors backend middleware. Session list / revoke to be completed in build.

5 Functional requirements

FR-1 — Staff auth & employee management

- Email/password login, OAuth hooks (IDs configurable), JWT sessions, rate-limited auth.

- Employee CRUD (**/api/users**), roles, departments, designation / joining fields, activate / deactivate, password reset.
- Permission-matrix UI; session list + revoke (to be completed).

FR-2 — Patient management

Registration (demographics, contact, referral doctor / agent), global search, and patient-wise case / bill / report history.

FR-3 — Billing & cases

- New bill with test / panel / package picker (search + autocomplete), rates, discounts, paid / partial / pending / due states, invoice numbering.
- Collect payment (full / partial, payment modes), due list, void-with-reason, bill barcode (SVG), invoice print.
- Cashbook (**GET /api/bills/cashbook**) with filters + date-range summary; expenses module.

FR-4 — Test master

- Test database: code / name / department / rate, normal ranges (**normalLow/High**, critical limits, age/sex applicability, unit, method), **tatHours**, derived-test flag + formula, interpretations.
- Categories, panels, packages; bulk rate revision (single-rate update exists; bulk flow to be completed).

FR-5 — Report lifecycle (pathology)

- Reports auto-generated from bills; Today / Due / Pending worklist with search, department filter, TAT countdown + due-soon badges, range-flagged result entry (HIGH / LOW / CRITICAL).
- Pathologist flow: sign (**POST /api/reports/:id/sign**), verify / reject-with-reason / request-changes + comments (outstanding endpoints delivered within the build), unsigned list, **/lab/verify** screen.
- Print/PDF preview from a **single template** (preview = print), QR block (**qrDataUrl** + **verificationUrl**, public route **/r/:token**), branded letterhead from lab profile.
- Barcode stickers + case / sample labels (bill barcode live; case / sample endpoints + sticker sheet to be completed).

FR-6 — Radiology (USG / X-ray)

- USG cases + my-cases; X-ray cases + my-cases: create, findings entry, drag-drop image upload with progress / retry, print preview, sign-off.
- Upload endpoints live (**POST /api/usg…/images**, **/api/xray…/images**); delete-image + image-in-PDF to be completed.

FR-7 — Signatures

Signature CRUD + department assignment (**/api/signatures**); manager UI + auto-selection by department at sign time; signatures rendered in PDF at print resolution.

FR-8 — Turnaround time (TAT)

Per-report and per-test **tatHours**; default + emergency / urgent TAT configuration screen (to be completed); due / overdue highlighting on a single lab-local-time convention.

FR-9 — Business & analytics

Accounts / ledger, due reports, department filter, referral / agent tracking, daily business grid, test-usage analytics, data export (JSON live; CSV + history to be completed).

FR-10 — Doctors, agents, settings

Doctor master + referral linkage; agent / collector tracking; lab-profile setup; registration-number config (**PPL-** default, prefix / format configurable); branding / white-labelling; SMS template screen (live).

FR-11 — Doctor portal / booking / support (minimal, as-is)

Existing minimal doctor portal, appointments / booking, and support ticketing (**/api/tickets**) are retained. No slot-availability engine and no advanced portal build-out in this scope.

FR-12 — Audit / jobs

Activity / audit-log viewer and failed-send retry dashboard (to be completed once capture middleware and job APIs land).

6 Patient portal (OTP login)

> **Flow**
>
> **Mobile Number** → **OTP (MSG91)** → **Verify** → **Report list** → **View / Download PDF.** No patient passwords. OTP expiry 5 min, max 3 attempts, rate-limited resend. Bills and reports also verifiable via token URLs + QR scan.

- OTP sessions are modelled in **OtpSession**; development and testing proceed on test credentials pending live MSG91 and DLT cutover.
- QR on every final report encodes the verification URL (not internal IDs) and survives PDF rendering.

7 Notifications

- Generic send API (**POST /api/notify/send**) + per-report send modal (patient / doctors / cc / bcc), single + bulk, template-driven SMS / Email.
- Delivery status (Sent / Failed / Retry) + per-report history (to be completed); failed sends never display false success.
- **OTP/SMS provider: MSG91** — wallet recharge and API key are client-provided; Sender ID and DLT template are wired by the build.
- **WhatsApp channel: excluded** from this phase (no WABA work). The **whatsappReportsEnabled** toggle remains disabled by default; WhatsApp messaging may be taken up as a Phase-2 enhancement.

8 Non-functional requirements

ID	Requirement
NFR-1	Responsive UI; fully usable at 1366 px desktop + mobile patient portal
NFR-2	JWT-secured APIs; server-side role checks; no passwords exposed; rate limits on auth / OTP
NFR-3	Preview layout == print / PDF layout (single template); QR survives PDF render
NFR-4	Idempotent verify / sign (double-click safe); failed sends never show false success
NFR-5	Single lab-local timezone convention for all TAT / due computation
NFR-6	Basic hygiene for public pages (Lighthouse snapshot, SEO meta)

9 Deployment & hosting (client-borne, separate)

Excluded from the Rs 60,000 build price — paid by the client at actuals:

Item	Plan
Hosting	**Vercel** — deployed from this repo (**vercel.json**). Client owns the account; any Pro usage / bandwidth / function overages billed to client
Database	**MongoDB Atlas Free (M0)** at launch. Storage / RAM / backup limits accepted; upgrades are the client's cost
Domain	**New registration (lump sum, client-borne)** — the domain is owned by the client; DNS connection to Vercel with sender setup is included
TLS	Via Vercel (included with platform)

Item	Plan
Backups / uploads	Per Atlas tier + host storage; persistent-storage upgrades if needed are extra

10 Commercials — Rs 60,000 fixed

Total fixed build cost: Rs 60,000 (fixed price). Covers design, development, integration wiring, QA, deployment to the client's Vercel + Atlas Free, and handover / training for §5–§7.

#	Head	Amount (Rs)	Covers
1	**Backend team**	**25,000**	REST APIs, auth/JWT + OTP sessions, billing & accounts, report lifecycle + sign/verify, USG/X-ray cases, uploads + PDF/QR/barcode, notify service + MSG91 wiring, roles, seed + test fixes
2	**Frontend team**	**25,000**	All screens & flows (registration, billing, worklists, result entry, verify, print preview, labels, cashbook, employees, settings/branding, portals), responsive polish, template–print parity
3	**Other: QA, integration & handover**	**10,000**	E2E testing + bug-fix pass, MSG91/Email wiring, Vercel + Atlas + domain connect, one-shot master import, 1 training session + handover docs
	Total	**60,000**	

Payment schedule (subject to acceptance)

40% advance (Rs 24,000) → 40% on UAT build (Rs 24,000) → 20% on go-live (Rs 12,000). Extra requests beyond this SRS are separately estimated.

The Rs 60,000 does **not** include §11 recurring / third-party costs or §12 out-of-scope items.

11 Recurring / third-party costs (not included)

Item	Borne by	Basis (Oct 2026)
Vercel overages	Client	Client-owned account; team deploys, client pays platform
Atlas beyond Free	Client	Free M0 now; upgrade if limits hit
Domain (lump sum)	Client	Registration + yearly renewal; client owns it
OTP / SMS — MSG91	Client (pay-as-you-go)	Per msg91.com/in/pricing/otp: **5,000 OTPs @ ~Rs 0.25 (~Rs 1,250 + 18% GST)**; ~Rs 0.22 @ 15k; ~Rs 0.20 @ 27k; down to **~Rs 0.18** at volume. At ~1,000/mo the starter pack lasts ~5 months. Client recharges wallet directly
DLT / Sender ID	Client	Regulatory; client registers, team wires approved template / ID
WhatsApp	Excluded	Excluded from this phase; may be taken up as a Phase-2 enhancement
Email delivery	Client	Client SMTP or provider account
Design / migration beyond one-shot masters	Extra	Quoted separately

12 Out of scope

- Multi-branch / SaaS tenancy, subscriptions, paywalls, trials.
- Inventory & stock, HR / payroll, instrument interfacing (HL7 / ASTM).
- Phlebotomist mobile app; native Android / iOS apps.
- Advanced analytics / BI; payment-gateway collection.
- Regional-language report templates (English-only scope); WhatsApp / WABA channel (excluded).

13 Assumptions & dependencies

- Single lab, single brand / letterhead; one Vercel project + one Atlas cluster, both held in the client's name.
- The client provides: lab profile artwork and content, test masters / rates, doctor / agent lists, MSG91 wallet + key (~1,000 OTPs/month), and SMTP access. DLT registration and Sender ID approval are pending client action; live SMS / OTP deployment is subject to their completion.
- Domain: new registration, client-borne (name to be finalised).
- Migration source is Labsmart exports: one-shot **masters** import is included; full history migration shall be estimated separately after export review.
- Staff list and signatories are to be confirmed prior to UAT (interim provision: up to 10 logins, 1–2 signatories); printing on **A4 with configurable sticker sizes**.
- English-only reports; free-tier limits accepted at launch; 30-day defect-fix window; UAT in a maximum of two consolidated rounds.

14 Acceptance criteria (sign-off checklist)

#	Check
A1	Staff login → register patient → bill → enter results → sign → print/PDF with QR + letterhead, end-to-end
A2	Patient Mobile + OTP login → report list → view / download (live MSG91 credits)
A3	Billing: collect / partial / due / void + cashbook totals reconcile for a test date range
A4	USG + X-ray: create case → upload image → preview → sign
A5	Bill barcode + QR-scan verification; sticker / label sheet on approved stock size
A6	Deployed on client Vercel + Atlas Free + domain connected; admin handover + training complete

15 Agreed terms & client dependencies

15A — Record of agreed terms (SRS v1.1, 03 October 2026)

Item	Agreed position	Implication
Domain	New registration	Two to three candidate names to be shortlisted; registration and renewal are client-borne; DNS connection included
Vercel + Atlas	Client-owned accounts	Deployment included; platform overages, if any, are client-borne
MSG91 volume	Approx. 1,000 / month	5,000-OTP starter pack (~Rs 1,250 + GST) covers roughly five months; wallet recharge and API key are client-provided
DLT + Sender ID	Pending (client action)	Live SMS / OTP deployment is subject to DLT completion; development proceeds on test credentials in parallel
WhatsApp	Excluded	Excluded from this phase; no associated cost or timeline

Item	Agreed position	Implication
Language	English only	Single-language report scope
Staff	To be confirmed	Interim provision: up to 10 logins with 1–2 signatories; confirmation required before UAT
Migration	Labsmart exports	Masters import included; full history migration estimated separately after CSV review
Tax treatment	Not applicable	Rs 60,000 fixed price
Milestones	For acceptance	Proposed 24k / 24k / 12k schedule, subject to acceptance
Printers	A4 + custom sticker stock	Reports on A4; configurable label layouts subject to stock approval
Defect-fix window	30 days post go-live	Defect fixes against this SRS for 30 days following go-live

15B — Client dependencies (inputs required)

#	Input	Dependent deliverable
1	Lab identity: name, address, phones, reg. no., timings, disclaimer + logo file	Letterhead / PDF / QR page
2	Two to three preferred domain names	Registration + go-live URLs
3	Labsmart CSV exports (tests / rates, doctors, patients, bills) + 2–3 sample report PDFs	Master import, migration estimate, template alignment
4	DLT registration started; Sender ID + template on approval	Live SMS / OTP
5	MSG91 account + starter recharge + API key	OTP wiring
6	Staff list + pathologist name(s) / title(s) / signature images	Logins + UAT + sign-off
7	Acceptance of the payment schedule (or an agreed alternative)	Work commencement / billing

Requirements outside §5 shall be handled under change control and estimated separately.

16 Support (AMC) & risk notes

AMC — Basic: Rs 24,000 per annum (Rs 2,000 per month)

Effective after the 30-day defect-fix window. **Includes:** defect fixes against this SRS, platform and dependency upkeep, MSG91 re-configuration, and break-fix support. **Excludes:** new features or modules, third-party charges, and data entry or migration. Billing is monthly in advance; termination on 30 days' notice.

Delivery risks & mitigations

Risk	Mitigation
DLT dependency	Development proceeds on MSG91 test credentials; cutover to the approved Sender ID upon DLT completion; OTP login validated with test credentials in the interim
Labsmart export windows (bills ~31d, patients ~365d)	Import masters first; quote full history only after reviewing actual CSVs; multi-batch exports if needed
Free-tier limits (Vercel / Atlas)	Accepted at launch; monitor usage; upgrade path and cost owned by client

Risk	Mitigation
Content delays (logo, disclaimer, rates)	Interim branding placeholders in the initial build; content frozen before the second UAT round

Sign-off

Client	Development team
Name: _____	Name: _____
Signature / Date: _____	Signature / Date: _____

End of SRS v1.1 — Pure Path Lab • Rs 60,000 fixed-price build • Infrastructure, domain and messaging at actuals • WhatsApp channel excluded • 30-day defect-fix window • AMC Basic Rs 2,000/month.